

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION CONTROL, LOW-VISCOSITY NBR PROCESSING, AND NBR-BASED FUNCTIONAL ARTICLES

1. ABSTRACT

This report synthesizes four patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related NBR processing routes. The target is low-ACN NBR, generally understood as NBR with reduced acrylonitrile incorporation relative to conventional grades, where composition control is used to tune oil resistance, flexibility, and processability. The patent set spans two synthesis-oriented disclosures and two application/process disclosures. EP2473279B1 focuses on post-polymerization olefin metathesis of nitrile rubber followed by hydrogenation to generate low-molecular-weight, low-Mooney HNBR with improved processability; although not a direct low-ACN synthesis patent, it is highly relevant because it addresses the downstream processing challenge often associated with low-ACN feedstocks and low-viscosity NBR-derived materials. EP2561011B1 discloses a composite-material process in which a butadiene-acrylonitrile copolymer (NIPOL 1312V) is introduced as a nitrile resin base into a BUSS co-kneader with carbon nanotubes and graphene, demonstrating continuous compounding at defined temperatures and screw speed. US10443339B2 is an elastomeric-material patent centered on carbon-based nanoplatelets in bore-tool elastomers; it is only indirectly relevant to NBR because it explicitly references NBR/HNBR among candidate elastomers but does not disclose NBR synthesis. US20200299532A1 discloses antistatic glove manufacture using nitrile rubber/natural rubber blends, including multilayer latex dipping and chlorination, and provides a practical application context for nitrile rubber formulations where lower acrylonitrile content can be desirable for softness and processing balance. Across the set, the dominant trends are composition blending, continuous mixing/compounding, and property-driven formulation rather than direct monomer-feed synthesis. The evidence most strongly supports low-ACN NBR development through controlled butadiene/acrylonitrile copolymerization in emulsion, followed by optional metathesis/hydrogenation or downstream blending into functional articles and composites.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473279B1
- Patent Title: Metathesis of a nitrile rubber in the presence of a transition metal catalyst
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis of nitrile rubber in the presence of a transition metal catalyst, followed by hydrogenation to obtain low-molecular-weight hydrogenated nitrile rubber (HNBR). The process is carried out in monochlorobenzene, with 1-hexene used as a co-olefin in the examples, and the metathesized rubber is then hydrogenated under high hydrogen pressure. - 1-Hexene loading: 4 phr — Examples 5-6 - Reaction temperature: 22°C — Examples 5-6 - Metathesis catalyst loading: 0.06 phr tris-(triphenylphosphine) rhodium chloride — Examples 5-6 - Hydrogen pressure: 85 bar — Examples 5-6 - Hydrogenation temperature: 138°C — Examples 5-6 - Agitation speed: 600 rpm — Examples 5-6 - Hydrogenation time: 4 h — Examples 5-6 - Hydrogenation level: <0.9% — Examples 5-6 - Metathesis catalyst loading: 0.1 phr — Example 7 - Solvent content: 1,300 ppm — Example 7 - Mn: 19,000 g/mol — Example 7 - Mw: 34,000 g/mol — Example 7 - PDI: 1.78 — Example 7 - Control Mn: 69,000 g/mol — Example 7 - Control Mw: 217,500 g/mol — Example 7 - Control PDI: 3.15 — Example 7

Relevant Experimental Evidence - Example 1: Increasing catalyst level changed molecular-weight metrics from Mn 24,500 to 15,000 g/mol and Mw 48,000 to 25,500 g/mol across the series. - Example 2: Similar catalyst variation gave Mn 19,000 g/mol and Mw 35,500 g/mol, showing molecular-weight reduction by metathesis. - Example 3: Further catalyst increase produced Mn 16,000 g/mol and Mw 29,500 g/mol, again lowering molecular weight. - Example 4: At 0.1 phr catalyst, Mn 15,000 g/mol and Mw 25,500 g/mol were reported, demonstrating strong chain scission. - Example 7: Using Perbunan T 3429, metathesis reduced Mw to 34,000 g/mol and PDI to 1.78 versus the control Mw 217,500 g/mol and PDI 3.15.

Technical Relevance This patent is relevant because low-acrylonitrile NBR grades often require improved processability, and this disclosure provides a route to low-Mw, low-PDI nitrile rubber/HNBR derived from NBR. While it does not change ACN content directly, it is important for making low-ACN NBR-derived materials easier to compound and use.

Patent 2

- Patent Number: EP2561011B1
- Patent Title: Thermoplastic and/or elastomeric composite material based on carbon nanotubes
- ...

- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses continuous compounding of a nitrile resin base with carbon nanotubes and graphene in a BUSS co-kneader, followed by granulation. In Example 7, a butadiene-acrylonitrile copolymer is preheated and injected as a liquid into the co-kneader together with nanotubes, graphene, and an acrylic copolymer powder to form a composite material. - Co-kneader model: BUSS MDK 46 — Example 7 - L/D: 11 — Example 7 - Nitrile resin preheat temperature: 160°C — Example 7 - Injection temperature: 190°C — Example 7 - Co-kneader temperature: 200°C — Example 7 - Flow rate: 12 kg/h — Example 7 - Screw speed: 240 rpm — Example 7 - Composite composition: 35% by weight CNT — Example 7 - Composite composition: 5% by weight graphene — Example 7 - Composite composition: 55% by weight nitrile resin — Example 7 - Composite composition: 5% by weight acrylic copolymer — Example 7 - Drying temperature: about 50°C — Example 7 - Zone 1/Zone 2 temperature: 280/290°C — Example 1 - Flow rate: 13 kg/h — Example 1 - Twin-screw extruder diameter: 16 mm — Example 2 - Twin-screw extruder L/D: 25 — Example 2 - Process window: 230-290°C — Example 2 - Zone 1/Zone 2 temperature: 290/290°C — Example 3 - Flow rate: 11 kg/h — Example 3 - Flow rate reduction: 10 kg/h — Example 3 - Temperature in last zone: 320°C — Example 3 - Temperature in last zone: 300°C — Example 3 - Flow rate: 15 kg/h — Example 5 - Production duration: about 40 h — Example 5 - Internal mixer fill factor: 74% — Example 56 - Rotor speed: 60 rpm — Example 56 - Initial temperature: 60°C — Example 56 - Vulcanization temperature: 180°C — Example 56 - Vulcanization time: 15 min — Example 56 - Injection moulding fill pressure: 50 bar — Example 56 - Injection moulding fill pressure: 97 bar — Example 56 - Internal temperature: 100°C — Example 56 - Cure temperature: 190°C — Example 56

Relevant Experimental Evidence - Example 1: CNT/graphene/polyamide 12 formulations showed lower viscosity and lower power consumption when graphene was included. - Example 2: The graphene-containing formulations achieved electrical conduction in the 230-290°C process window at lower temperature than CNT-only controls. - Example 3: In PA-6, graphene reduced viscosity, stabilized production, and prevented wall fouling compared with CNT-only formulation. - Example 4: The inventive composite reduced the manufacturing temperature of the final product by 20°C while maintaining electrostatic dissipation. - Example 7: A nitrile-resin-based composite was produced with 35% CNT, 5% graphene, 55% nitrile resin, and 5% acrylic copolymer at 200°C and 12 kg/h.

Technical Relevance This patent is relevant because it explicitly uses a butadiene-acrylonitrile copolymer as a nitrile resin base in a continuous compounding process. It supports the use of NBR-derived materials in functional composites and provides process conditions useful for handling low-viscosity or composition-tuned nitrile rubber systems.

Patent 3

- Patent Number: US10443339B2
- Patent Title: Elastomeric materials
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR synthesis; it describes elastomeric bore-tool components containing carbon-based nanoplatelets in an elastomeric matrix. The specification discusses elastomers such as NBR and HNBR as candidate materials for high-temperature service, but the invention is centered on reinforcement and downhole article design rather than polymerization. - Carbon-based nanoplatelets content: less than approximately 10 percent by weight — Claim 1 - HPHT environment pressure: up to about 138 MPa — Detailed description - HPHT environment temperature: up to about 205°C — Detailed description - Ultra-HPHT environment pressure: up to about 241 MPa — Detailed description - Ultra-HPHT environment temperature: up to about 260°C — Detailed description - HNBR mechanical properties decrease at about 160°C — Detailed description - AFLAS mechanical properties decrease at about 220°C — Detailed description - Elastomer modulus: about 3 MPa — Detailed description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it identifies NBR/HNBR as useful elastomers for harsh environments. It does not advance low-ACN synthesis, but it helps define application requirements that low-ACN NBR formulations may need to meet.

Patent 4

- Patent Number: US20200299532A1
- Patent Title: Antistatic gloves and process for making same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses antistatic glove manufacture using nitrile rubber/natural rubber latex blends, including multilayer dipping, drying, leaching, chlorination, and curing. The process uses a first latex composition richer in nitrile rubber for the outer layer and a second latex composition richer in natural rubber for the inner layer to balance antistatic performance and comfort. - Relative humidity: $12\pm3\%$ — Example 1 - Temperature: $22\pm2^{\circ}\text{C}$ — Example 1 - Surface resistivity criterion: below about $10^{14} \Omega/\text{sq}$ — Abstract - Static decay time criterion: less than about 60 s — Abstract - Former heating temperature: 70°C — Example 2 - Former temperature: $59\text{--}66^{\circ}\text{C}$ — Example 2 - Coagulant temperature: $53\text{--}59^{\circ}\text{C}$ — Example 2 - Drying temperature: 70°C — Example 2 - Drying time: 3-5 min — Example 2 - First latex dwell time: 12-18 s — Example 2 - Intermediate drying temperature: 135°C — Example 2 - Intermediate drying time: 1 min — Example 2 - Second latex dwell time: 5-10 s — Example 2 - Leaching temperature: 40°C — Example 2 - Leaching time: 3 min — Example 2 - Final curing temperature: 135°C — Example 2 - Final curing time: 20 min — Example 2 - Chlorine solution concentration: about 300 ppm — Example 2 - Chlorination time: 35 min — Example 2 - Outer layer composition: 95% nitrile rubber latex / 5% natural rubber latex — Example 3 - Inner layer composition: 5% nitrile rubber latex / 95% natural rubber latex — Example 3 - Surface resistivity: $3.3 \times 10^{12} \Omega/\text{sq}$ — Example 3 - Static decay time: 21.8 s — Example 3 - Modulus at 500% elongation: 3.6 MPa — Example 3 - Ultimate elongation: 609% — Example 3 - Outer layer composition: 80% nitrile rubber latex / 20% natural rubber latex — Example 4 - Inner layer composition: 20% nitrile rubber latex / 80% natural rubber latex — Example 4 - Surface resistivity: $1.7 \times 10^{13} \Omega/\text{sq}$ — Example 4 - Static decay time: 27.2 s — Example 4 - Modulus at 500% elongation: 3.7 MPa — Example 4 - Ultimate elongation: 569% — Example 4 - Outer layer composition: 60% nitrile rubber latex / 40% natural rubber latex — Example 5 - Inner layer composition: 40% nitrile rubber latex / 60% natural rubber latex — Example 5 - Surface resistivity: $2.8 \times 10^{13} \Omega/\text{sq}$ — Example 5 - Static decay time: 29.2 s — Example 5 - Modulus at 500% elongation: 4.2 MPa — Example 5 - Ultimate elongation: 569% — Example 5 - Outer layer composition: 51% nitrile rubber latex / 49% natural rubber latex — Example 6 - Inner layer composition: 49% nitrile rubber latex / 51% natural rubber latex — Example 6 - Surface resistivity: $3.0 \times 10^{13} \Omega/\text{sq}$ — Example 6 - Static decay time: 34.0 s — Example 6 - Modulus at 500% elongation: 6.6 MPa — Example 6 - Ultimate elongation: 569% — Example 6

Relevant Experimental Evidence - Example 1: Commercial nitrile gloves and a nitrile/natural rubber laminate were compared for static resistance under controlled humidity and temperature. - Example 2: A double-dip process was used to make a two-layer glove with nitrile-rich outer latex and natural-rubber-rich inner latex. - Example 3: A 95/5 nitrile/natural rubber outer layer and 5/95 inner layer gave antistatic performance with no immediate delamination after curing. - Example 4: An 80/20 nitrile/natural rubber outer layer and 20/80 inner layer maintained antistatic behavior and improved interlayer stability. - Example 6: A 51/49 nitrile/natural rubber laminate still showed antistatic properties, with higher modulus at 500% elongation.

Technical Relevance This patent is relevant because it uses nitrile rubber latex in practical antistatic articles and shows how nitrile-rich formulations can be blended and processed into multilayer products. Although it does not disclose NBR synthesis, it is useful for understanding downstream formulation targets for low-ACN nitrile rubber in gloves and similar articles.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473279B1 is the only patent in the set that directly addresses nitrile-rubber molecular-weight reduction by metathesis/hydrogenation; the others are application or compounding disclosures rather than direct NBR synthesis.
- EP2561011B1 and US20200299532A1 both use nitrile rubber in formulated products, but EP2561011B1 emphasizes continuous high-shear compounding with CNT/graphene, whereas US20200299532A1 emphasizes latex blending, dipping, chlorination, and curing for antistatic gloves.
- US10443339B2 is the least synthesis-relevant to low-ACN NBR; it mentions NBR/HNBR as candidate elastomers for high-temperature bore tools but focuses on nanoplatelet reinforcement and service environment rather than monomer ratio or polymerization.
- Only EP2561011B1 explicitly discloses a butadiene-acrylonitrile copolymer in a process context, with a nitrile resin base injected at 190°C and compounded at 200°C, while US20200299532A1 uses nitrile rubber latex blends with natural rubber to tune comfort and antistatic behavior.
- The disclosed process temperatures span from latex dipping conditions near 40-135°C in US20200299532A1 to 160-200°C compounding in EP2561011B1 and 138°C hydrogenation in EP2473279B1, showing that the patent set covers both low-temperature latex processing and higher-temperature melt/solution processing.
- EP2473279B1 provides the strongest evidence for low-viscosity nitrile-rubber processing, with Mw reduced to 34,000 g/mol and PDI to 1.78 in Example 7, which is directly useful for improving processability of low-ACN NBR-derived materials.
- US20200299532A1 shows that nitrile content can be reduced in multilayer glove structures by blending nitrile rubber with natural rubber, but it does not disclose acrylonitrile monomer content or synthesis feed ratios.
- Across the set, no patent explicitly discloses a low-acrylonitrile monomer feed ratio for NBR polymerization; the evidence is strongest for downstream composition control and processability rather than direct low-ACN copolymer synthesis.

4. CONCLUSION

The patent evidence does not provide a direct low-acrylonitrile NBR polymerization recipe, but it does identify the most relevant process directions for making and using low-ACN nitrile rubber materials. The strongest synthesis-adjacent guidance comes from EP2473279B1, which demonstrates metathesis/hydrogenation to lower molecular weight and improve processability, and from EP2561011B1, which shows that nitrile rubber can be continuously compounded with fillers and functional additives at controlled temperatures and flow rates. US20200299532A1 confirms that nitrile-rich latex can be blended with natural rubber to tune comfort, antistatic

behavior, and interlayer adhesion in finished articles. Based on the evidence, the most defensible recommended parameters for low-ACN NBR development are controlled butadiene/acrylonitrile copolymerization followed by careful molecular-weight management, with downstream processing in the approximate 160-200°C range for compounding or 40-135°C for latex article formation, depending on the product form. For low-viscosity, easily processable nitrile rubber, the evidence supports targeting low Mw and narrow PDI rather than relying on unsubstantiated changes in acrylonitrile content alone.

5. REFERENCES

- EP2473279B1 | Metathesis of a nitrile rubber in the presence of a transition metal catalyst | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473279B1/en>
- EP2561011B1 | Thermoplastic and/or elastomeric composite material based on carbon nanotubes ... | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2561011B1/en>
- US10443339B2 | Elastomeric materials | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10443339B2/en>
- US20200299532A1 | Antistatic gloves and process for making same | Not disclosed | US | 2020 | <https://patents.google.com/patent/US20200299532A1/en>